Roll no.: 24BCB120D

Name: Takshil Patel

LAB SESSION 5: ARCHITECTURE DESIGN

1. Architecture Type

- **Model-View-Controller (MVC) Pattern**
  - **Model:** MySQL database containing Users, Books, Borrow Records, Reservations, Notifications and Fines.
  - **View:** React.js web application for login, book search, borrow/return, reservation, dashboard and reports.
  - **Controller:** Node.js + Express.js REST API handling authentication, business logic, API requests and validation.

2. Component Diagram

- **Components:**
  - **Authentication Module:** Handles user login and authentication.
  - **Book Management Module:** Handles book search, details, availability and management.
  - **Borrow/Return Module:** Handles borrowing, returning and renewal.
  - **Reservation Module:** Handles book reservations and cancellation.
  - **Fine Management Module:** Calculates and manages fines.
  - **File Management Module:** Handles library files.
  - **Notification Module:** Handles reminders and notifications.
  - **Reports & Analytics Module:** Generates library reports.
- **Database:** MySQL DB.
  - Stores users, books, borrow records, reservations, notifications and fines.
- **Connections:**
  - Backend Modules → MySQL DB: Store and retrieve library data.
  - React.js → Node.js/Express: Sends REST API requests.
  - Node.js/Express → React.js: Returns JSON responses.

3. Deployment Diagram

- **Node 1 (Client Device):** Student / Librarian / Admin device.
  - **Component:** Web Browser / React.js.
  - **Purpose:** Provides the user interface.
- **Node 2 (Application Server):** Node.js + Express.js.

    - **Components:** REST API and backend modules.
    - **Purpose:** Processes requests and executes library operations.
- **Node 3 (Database Server):** MySQL Server.
    - **Component:** MySQL Database.
    - **Stores:** Users, Books, Borrow Records, Reservations, Notifications and Fines.
- **Communication:**
    - Client ↔ Application Server: **HTTP/HTTPS**
    - Application Server ↔ Database Server: **SQL queries and results**.

